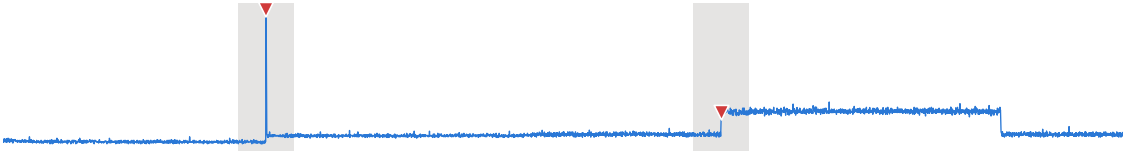
Sentinel a live anomaly detector that explains every alarm



How to read. a real database server's CPU, streamed one value at a time. Grey bands: the two moments experts labelled as anomalies. ▼ Sentinel's alarms: both anomalies caught, nothing else flagged.

The problem

Machines and apps send a number every few minutes. Flag the truly abnormal moments **live**, without false alarms, and say **why**.

Our solution

Five watchers score each new value; a **rarity meter** learned per signal turns it into a probability. Alarm only if rarer than **1 in 100,000**.

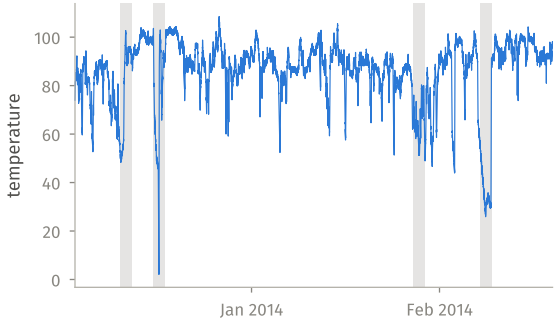
The result

All 58 benchmark signals: **event-F1 0.594**, the **best of 8 methods** tested (next: 0.316), **39** false alarms in 365,558 values, **61 µs** per value.

if/elif/else Aarohi Gattewar · Phalak Mehta · Harshit Singh

Every chart in this deck has a "How to read" note.

The problem: catch real failures in live data, without crying wolf



How to read. the temperature inside an industrial machine over three months. Grey bands: the 4 moments experts labelled as anomalies, around a known system failure. A good detector raises an alarm inside each band and nowhere else.

What the challenge asks

Input	a stream of values, one at a time. Decide at once, never look ahead.
Output	for every value: an anomaly score, a NORMAL / ANOMALOUS label, and evidence explaining each alarm.
Scored by	event-F1 (main score), precision, recall, detection delay, runtime.

Two ways to fail

Stay quiet: the early warning is missed and the failure becomes an outage.

Cry wolf: too many alarms, so people stop listening and miss the real one.

Why it matters

Cloud servers, factories, online ads, city traffic, social media: a missed failure costs money, a noisy alarm costs trust.

Our goal. Alarm only when a value is truly unusual *for its own signal*, decide live, and explain every alarm in one sentence.

Why it is hard: every signal is different, and extremes are normal

The data (Numenta benchmark)

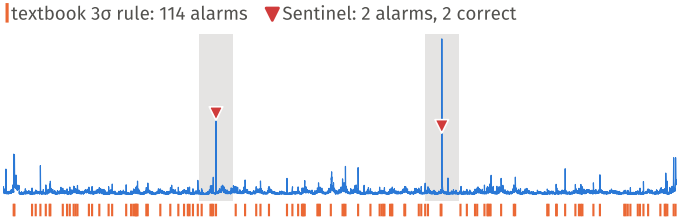
	signals	values	anomalies
Twitter mentions	10	158,631	33
Cloud servers	17	67,740	30
Known failures	7	69,561	19
Road traffic	7	15,664	14
Online ad prices	6	9,610	14
Synthetic	11	44,352	6
Total	**58**	**365,558**	**116**

What the data taught us

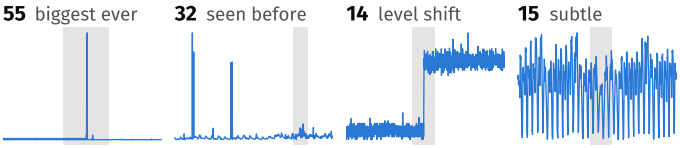
Extremes are normal: 47 of 58 signals have heavy tails.

Rhythm matters: 18 signals repeat every day.

Preprocessing: none needed. No missing values; read in time order, unscaled.



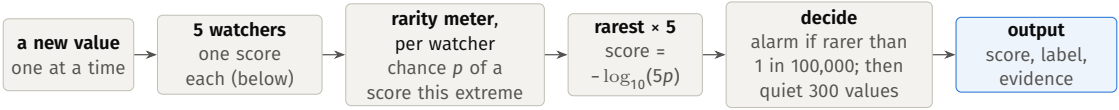
How to read. tweets about Facebook. Orange ticks: the textbook 3σ rule fires 114 times. ▼ Sentinel fires twice, both inside labelled anomalies (grey bands).



How to read. the 116 anomalies by type. “Seen before”: a bigger spike came earlier.

What this means. The 3σ rule raises **2,403 alarms for 116 real anomalies**, so Sentinel alarms on rarity instead.

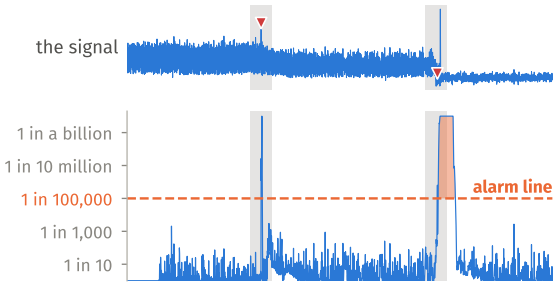
Our solution: five watchers ask one question, “how rare is this?”



watcher	what it measures, for every value	looks back (past only)	what it catches, with a real alarm it wrote	alarms (real)
Spike	distance from its running average, in running spreads (outliers clipped)	≈ 50 values	spikes and dips: *“Value 50.00 is 99.4 scales above the learned baseline 0.32”*	43 (30)
Routine	the same, for the change since a day earlier; daily signals only	one day	a normal value at the wrong time: *“Value 40.00 breaks the repeating pattern”*	7 (6)
New level	a fast average minus a slow average, in spreads	10 vs. 200	a level that moves and stays: *“recent mean 6.40 vs baseline 0.43”* (tweets)	24 (12)
Jitter	recent step size divided by long-run step size (log)	10 vs. 200	noise bursts or a frozen sensor: *“Signal went unusually flat: volatility is 0.21x its baseline”*	9 (3)
New shape	distance from the last 8 values to their closest match in the past 2,000	2,000 values	a never-seen pattern (repeats stop alarming): *“closest past match differs by 0.27 per point”*	23 (16)

How to read. One row per watcher. “Alarms (real)”: of the 106 final alarms, how many it raised and how many were real. Each value is scored *before* the watchers learn from it.

The key idea: alarm on rarity, not on a fixed number



How to read. top, a cloud server's CPU. Bottom, each value's rarity *for this signal*, as "1 in N". Only values above the dashed line alarm: two here, both real.

How the rarity meter works. Each watcher fits a curve (extreme-value theory) to its largest 5% of past scores; it gives p, the chance of a score this large. Score = $-\log_{10}(5 \cdot \min p)$; alarm if ≥ 5.

Why this model. No training set: it learns each signal from its first 300 values. Unlike an autoencoder, it keeps adapting and explains every alarm.

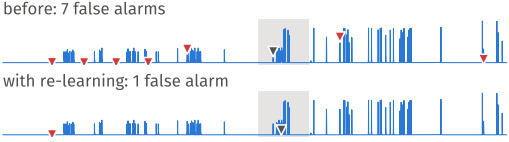
Two fixes we made this round

Fix 1 · Don't jump to conclusions. After warm-up a watcher has seen only about 12 extremes; its curve was often far too heavy ("huge values are normal"), so obvious spikes could not alarm. Now a heavy tail needs real evidence.



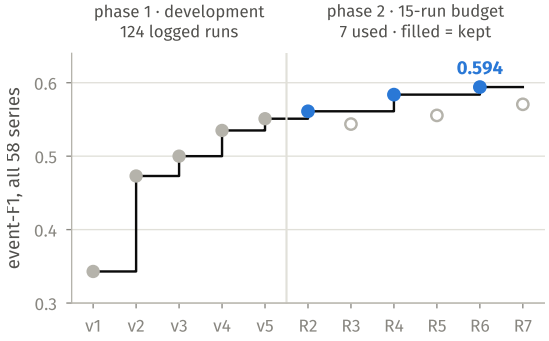
A real spike in an hourly ad price. Biggest-ever spikes caught: **31 → 37**.

Fix 2 · Accept a new normal. If a problem is still on after the 300-value quiet period, the signal has changed: Sentinel re-learns normal instead of alarming again and again.



Disk writes: false alarms **7 → 1**, no anomaly lost.

How we tuned it: one idea per run, kept only if the score went up



How to read. each dot is one full run on all 58 signals; the line is the best score so far. Grey: development versions v1 to v5. Blue: budgeted runs, filled if the idea was kept, hollow if dropped.

run	what changed	event-F1	decision
1	starting point (v5), reproduced	0.551	reference
2	fix 1, light	0.561	**kept**
3	fix 1, strong	0.544	dropped
4	+ fix 2 (re-learning)	0.584	**kept**
5	shorter learning: 300 → 200 values	0.556	dropped
6	fix 1, medium + re-learning	**0.594**	**kept, final**
7	fix 1, strong + re-learning	0.571	dropped

Search strategy

Budget: at most 15 runs; one run = all 58 signals, scored once.

Think first: ideas were tested on saved outputs for free; 7 were dropped without a run.

One change per run, its hypothesis written down before it started.

Keep rule: event-F1 must rise *and* more categories must gain than lose.

What this means. 0.551 → **0.594** with 7 of the 15 runs (8 kept in reserve). Only global settings were changed, never one signal at a time.

How we measure it: per incident, the way the challenge scores it

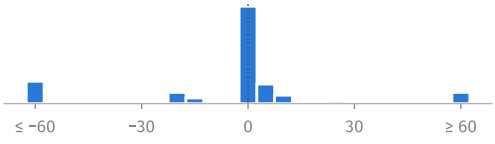


How to read. grey bands are labelled anomalies. Alarm in a band: caught. Alarm outside every band: false alarm. Band with no alarm: missed. One burst of alarms counts once.

metric	plain question	ours
Precision	of our alarms, how many were real?	0.63
Recall	of the real anomalies, how many were caught?	0.56
Event-F1	both combined: the main score	**0.594**
Delay	first alarm minus labelled anomaly point	0 steps

Checks. One setting for all signals · a category left out: still **0.594** · 5 clean signals: **0 alarms**.

category	anomalies	precision	recall	event-F1
Synthetic	6	1.00	0.83	0.91
Twitter mentions	33	0.57	0.76	0.65
Known failures	19	0.52	0.63	0.57
Cloud servers	30	0.67	0.47	0.55
Online ad prices	14	1.00	0.36	0.53
Road traffic	14	0.80	0.29	0.42
All	**116**	**0.63**	**0.56**	**0.594**



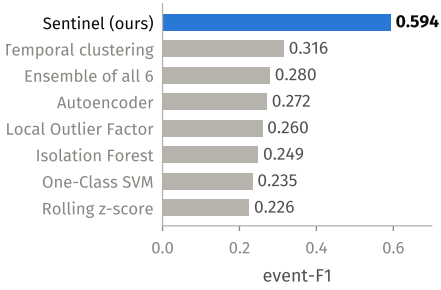
How to read. first alarm relative to the labelled anomaly point (0). Median **0 steps**; 66% within 10.

What this means. Right 2 times in 3 when it alarms, over half of all anomalies caught, on time.

Results: the best of all 8 methods, with 6 to 15 times fewer false alarms

method	event-F1	precision	recall	false alarms	µs/value
Sentinel (ours)	**0.594**	**0.63**	0.56	**39**	61
Temporal clustering	0.316	0.22	0.53	249	6
Ensemble of all 6	0.280	0.18	0.65	419	221
Autoencoder	0.272	0.18	0.60	390	140
Local Outlier Factor	0.260	0.16	0.62	457	13
Isolation Forest	0.249	0.18	0.42	261	54
One-Class SVM	0.235	0.14	0.68	594	8
Rolling z-score	0.226	0.15	0.47	351	0.5

How to read. every model family named in the problem statement, same data, same scoring, each learning from the first 300 values of every signal. Each got its **best threshold, chosen using the answers**; Sentinel uses one fixed setting. Sentinel: 65 of 116 anomalies caught; the others' higher recall costs 6 to 15× more false alarms.



Why 0.594 is a strong score

1.9× the runner-up

temporal clustering, even with its threshold tuned on the answers; 2.3× the best standard method.

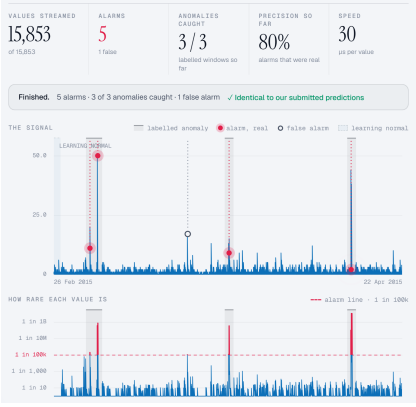
Few alarms, mostly real

one false alarm per 9,400 values, about one a month on a 5-minute signal.

Why papers quote 0.9+

they use "point adjustment", where random alarms score 0.92 (0.16 here).

See it work: every alarm comes with a reason a person can read



How to read. our live web demo after streaming 15,853 tweet counts about CVS. Red dots: real alarms; hollow dot: the one false alarm.

The evidence, exactly as Sentinel wrote it

SPIKE · 5 Mar 19:57 · score 8.5
Value 50.00 is 99.4 scales above the learned baseline 0.32 (normal band −1.18 to 1.82).

NEW LEVEL · 26 Mar 14:07 · score 5.1
Sustained level shift: recent mean 6.40 vs baseline 0.43 (8.2 scales above).

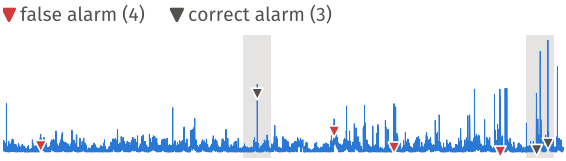
NEW SHAPE · 14 Apr 21:02 · score 6.0
The last 8 points form a pattern not seen before: the closest past match (1,750 steps ago) differs by 11.40 per point.

Sample predictions (predictions.csv, one row per value)

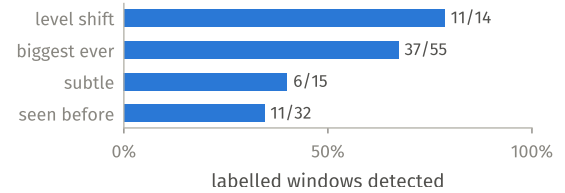
timestamp	value	score	threshold	label	watcher
2015-03-05 19:52:53	0.0	0.000	5.0	NORMAL	
2015-03-05 19:57:53	50.0	8.525	5.0	ANOMALOUS	level

What this means. An operator sees which watcher fired, how rare the value was and what normal looked like.

Where it still fails, why 0.594 is the ceiling for now, and the bottom line



How to read. tweets about IBM. Red triangles are false alarms: big bursts that nobody labelled. Most of our 39 false alarms look like this.



How to read. share of each anomaly type that Sentinel catches. “Seen before” spikes stay hard: rarity alone cannot tell them from the earlier, bigger one.

Why not higher than 0.594?

Hard by nature: 21 of the 51 misses are “seen before” spikes and 9 are subtle changes; flagging them would flag normal data too.

Every other idea cost more than it gained: a catch is worth about 2.4 false alarms in event-F1; our dropped runs paid 6 to 14. Tuning peaked at 0.594.

Warm-up: 3 anomalies start before any baseline exists.

Future work (needs data we do not have)

Operator feedback: needs people marking alarms in real use; reusing the answers would be cheating.

Related signals together: NAB gives each signal alone.

Holiday effects: no signal covers a full year, so each holiday appears at most once.

Bottom line. Sentinel finds 65 of 116 real anomalies, 63% of its alarms are correct, it runs live at 61 µs per value, and it explains every alarm.